

A product theorem for Δ -spaces with a point-finite-assignment factor

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Status: candidate_partial_result_likely_valid.

1 Source problem and later status

Kąkol and Leiderman ask whether the product of two compact Δ -spaces must again be a Δ -space [1, Problem 5.9, PDF p. 12]. (Later references use Problem 5.10, reflecting a numbering change in the journal version.)

$X \subset \mathbb{R}$ such that its square X^2 is not a Q -set, is consistent with ZFC.

Problem 5.8. *Is the existence of a Δ -set $X \subset \mathbb{R}$ such that its square X^2 is not a Δ -set, consistent with ZFC?*

It is known that the finite product of scattered Eberlein compact spaces is a scattered Eberlein compact.

Problem 5.9. *Let X be the product of two compact spaces X_1 and X_2 such that each X_i belongs to the class Δ . Does X belong to the class Δ ?*

Our last problem is inspired by Theorem 3.10.

Problem 5.10. *Let X be any scattered compact space with a finite scattered height. Does X belong to the class Δ ?*

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REFERENCES

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Leiderman and Szeptycki explicitly report that the question remains open even for compact factors [3, Section 5]. The result below gives a positive answer whenever one factor has a point-finite neighborhood assignment, and a countable-closed-cover extension reaches known non-Eberlein compact examples.

2 Definitions and idea of the proof

A Tychonoff space X is a Δ -space if every decreasing sequence of subsets with empty intersection admits a decreasing open expansion with empty intersection. We use the equivalent characterization from [1, Theorem 2.1]: every countable pairwise disjoint family of subsets of X admits a point-finite open expansion.

A *point-finite neighborhood assignment* (PFNA) on Y is a family $\{N_y : y \in Y\}$ such that N_y is an open neighborhood of y and, for each $t \in Y$, only finitely many y satisfy $t \in N_y$.

Idea of the proof

Given disjoint sets in $X \times Y$, freeze $y \in Y$ and expand their vertical sections point-finitely in X . Thicken each fiberwise expansion by the assigned neighborhood N_y and take the union over y . At a product point (x, t) , point-finiteness on Y leaves only finitely many relevant fibers; point-finiteness in X leaves only finitely many indices on each such fiber. This two-stage finiteness is exactly what the product problem normally lacks.

3 The fiberwise product theorem

Theorem 1 (PFNA factor theorem). *Let X be a Δ -space and let Y be a Tychonoff space admitting a point-finite neighborhood assignment. Then $X \times Y$ is a Δ -space.*

Proof. Fix a pairwise disjoint sequence $\{A_n : n \in \mathbb{N}\}$ of subsets of $X \times Y$. For $y \in Y$, let

$$A_n(y) = \{x \in X : (x, y) \in A_n\}.$$

For fixed y , the sets $A_n(y)$ are pairwise disjoint. Since X is a Δ -space, choose a point-finite family of open sets $\{U_{n,y} : n \in \mathbb{N}\}$ in X such that $A_n(y) \subseteq U_{n,y}$ for every n .

Let $\{N_y : y \in Y\}$ be a PFNA and define

$$W_n = \bigcup_{y \in Y} U_{n,y} \times N_y.$$

Each W_n is open. If $(x, y) \in A_n$, then $x \in U_{n,y}$ and $y \in N_y$, so $(x, y) \in W_n$.

It remains to check point-finiteness. Fix $(x, t) \in X \times Y$ and put

$$F_t = \{y \in Y : t \in N_y\}.$$

The set F_t is finite. Moreover,

$$\{n : (x, t) \in W_n\} \subseteq \bigcup_{y \in F_t} \{n : x \in U_{n,y}\}.$$

Every set on the right is finite by the chosen fiberwise expansion, and there are only finitely many of them. Thus $\{W_n : n \in \mathbb{N}\}$ is point-finite. The disjoint-family characterization now gives $X \times Y \in \Delta$. $\square$

Corollary 2. *Suppose $X \in \Delta$ and $Y = \bigcup_{m \in \mathbb{N}} Y_m$, where every Y_m is closed in Y and admits a PFNA. Then $X \times Y \in \Delta$.*

Proof. Theorem 1 gives $X \times Y_m \in \Delta$ for every m , and these subspaces are closed in $X \times Y$. Kąkol and Leiderman's countable closed-union theorem [2, Proposition 2.2] now yields the conclusion. $\square$

4 Compact consequences

Every scattered Eberlein compactum admits a PFNA [1, Remark 2.4 and the discussion after Theorem 3.7]; see also [3, Section 6]. Therefore:

Corollary 3. *Let X be any Δ -space.*

1. *If K is a scattered Eberlein compact space, then $X \times K \in \Delta$.*
2. *If a compact space K is a countable union of scattered Eberlein compact subspaces, then $X \times K \in \Delta$.*
3. *In particular, the conclusion holds for every compact scattered K of Cantor–Bendixson height at most two.*

Proof. Part (1) is Theorem 1. In (2), the compact subspaces are closed, so Corollary 2 applies. A compact scattered space of height at most two is a finite topological sum of one-point compactifications of discrete spaces, up to finitely many isolated points; hence it is a scattered Eberlein compactum. $\square$

The next case is useful because the factor need not itself be Eberlein. Let $\mathcal{A}$ be an almost disjoint family of infinite subsets of $\mathbb{N}$, let $\Psi(\mathcal{A}) = \mathbb{N} \cup \mathcal{A}$ be the corresponding Isbell–Mrówka space, and let $K(\mathcal{A})$ be its one-point compactification.

Corollary 4. *For every Δ -space X and every almost disjoint family $\mathcal{A} \subseteq [\mathbb{N}]^\omega$,*

$$X \times K(\mathcal{A}) \in \Delta.$$

When $\mathcal{A}$ is uncountable, this includes the non-Eberlein compacta used in [1, Theorem 3.10].

Proof. Inside $K(\mathcal{A})$, the set $K_0 = \mathcal{A} \cup \{\infty\}$ is closed and is the one-point compactification of the discrete space $\mathcal{A}$. Thus K_0 is a scattered Eberlein compactum and admits a PFNA. The remaining points are the countably many closed singletons $\{n\}$, $n \in \mathbb{N}$. Hence

$$K(\mathcal{A}) = K_0 \cup \bigcup_{n \in \mathbb{N}} \{n\}$$

is a countable closed PFNA cover, and Corollary 2 applies. $\square$

The same proof works for a generalized $\Psi(D, \mathcal{A})$ whenever $D = \bigcup_m D_m$ and every D_m is closed discrete in $\Psi(D, \mathcal{A})$: use the closed compact PFNA pieces $\mathcal{A} \cup \{\infty\}$ and $D_m \cup \{\infty\}$.

5 Upgrade audit, limits, and novelty

Eight focused upgrade routes were checked. The successful chain was: fiberwise PFNA factor theorem; scattered Eberlein specialization; countable closed PFNA covers; height-two compacta; compactified classical Ψ -spaces; and the generalized Ψ exhaustion above. Two routes did not close the full problem. First, proving that every compact Δ -space has the needed countable Eberlein cover is itself an open problem reiterated as Problem 5.6 in [3]. Second, the ZFC rank-three non- Δ compactum constructed

in [3, Corollary 4.6] did not yield an embedding into, or representation as, a product of two compact Δ -spaces.

Thus the source product problem remains open when neither factor has a PFNA or a countable closed PFNA cover. The result here is a substantial positive subcase, not a full solution.

For novelty, the run’s four lightweight indexes and bounded searches using the arXiv id, title, authors, “point-finite neighborhood assignment,” “product,” “ Δ -space,” and “scattered Eberlein compact” found [2, 3] but no explicit statement of Theorem 1. This is a candidate result awaiting expert review, not a definitive priority claim.

Human-review recommendation

The core proof is short and should be checked first at the quantifier switch from the fiberwise expansions to W_n . The compact corollaries then require only verification of the cited PFNA and countable-closed-union facts. A reviewer should also run a broader topology-database search for an earlier abstract product lemma under the terminology “scant cover.”

References

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